Drift-Aware Input-Adaptive Temperature Scaling for Improved Calibration of Transformer Models under Distribution Shift

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Abstract

Transformer-based models achieve high predictive accuracy in natural language processing tasks but often exhibit degraded confidence calibration under distribution shift. While post-hoc temperature scaling improves in-domain calibration, it applies a global correction and fails to account for input-specific distributional deviation. In this work, we propose a drift-aware input-adaptive temperature scaling framework that dynamically adjusts predictive confidence based on embedding-space drift quantified via Mahalanobis distance. Our approach leverages covariance-aware representation statistics to estimate distributional deviation and modulates temperature as a function of normalized drift magnitude. Experiments on AG News (in-domain) and Yahoo Answers (shifted domain) demonstrate that the proposed method reduces Expected Calibration Error (ECE) by over 50% compared to the baseline (0.042 → 0.019) and significantly outperforms static temperature scaling. Reliability diagrams and hyperparameter sensitivity analysis confirm stable and robust calibration improvements. The proposed method is lightweight, architecture-agnostic, and easily integrated into existing transformer-based classification systems, providing a practical solution for drift-aware confidence calibration in real-world NLP applications.

Unlike static temperature scaling, the proposed method introduces input-conditioned calibration sensitivity without retraining or additional model parameters.

Keywords

Drift-aware calibration, temperature scaling, distribution shift, Mahalanobis distance, transformer models, confidence calibration, uncertainty estimation, embedding-space drift, domain shift robustness, natural language processing

1 Introduction

Deep neural networks, particularly transformer-based architectures, have become the dominant paradigm in natural language processing (NLP). Models such as BERT and its variants achieve high predictive accuracy across classification, question answering, and generation tasks. However, high accuracy does not guarantee reliable uncertainty estimation. Modern neural networks are frequently miscalibrated, meaning that predicted confidence does not align with empirical correctness likelihood. Overconfident predictions can lead to harmful downstream decisions, especially in safety-critical, financial, and decision-support applications.

Post-hoc calibration methods aim to correct confidence estimates without retraining the model. Among them, temperature scaling (TS) is widely adopted due to its simplicity and effectiveness. TS rescales logits using a single global temperature parameter optimized on validation data. While TS often improves in-domain calibration, it assumes homogeneous uncertainty across all inputs. This assumption breaks under distribution shift, where inputs deviate from the training distribution in semantic or statistical structure.

Recent work has demonstrated that distribution shift is reflected in representation space geometry. Embedding vectors encode semantic structure, and deviations from training distribution can be quantified via distance-based measures. In particular, Mahalanobis distance provides a covariance-aware metric that captures correlated deviations across embedding dimensions. However, existing approaches primarily use such distances for out-of-distribution detection rather than adaptive calibration.

In this work, we hypothesize that confidence calibration should be conditioned on input-specific distributional deviation rather than globally applied. To this end, we propose a drift-aware input-adaptive temperature scaling framework that dynamically adjusts predictive confidence based on embedding-space drift. Drift is quantified using Mahalanobis distance computed over transformer CLS embeddings. Temperature is then modulated as a function of normalized drift magnitude.

The main contributions of this work are:

1. We introduce a covariance-aware Mahalanobis embedding drift estimator tailored to transformer representations.
2. We propose an input-adaptive temperature scaling mechanism that modulates confidence based on representation drift.
3. We demonstrate significant calibration improvements over both baseline and static temperature scaling.
4. We provide robustness analysis through reliability diagrams and hyperparameter sensitivity evaluation.
5. We analyse confidence behaviour under distribution shift, highlighting adaptive uncertainty modulation.

The proposed framework is lightweight, architecture-agnostic, and requires no retraining, making it practical for real-world NLP systems.

2 Related Work

2.1 Confidence Calibration

Confidence calibration aims to align predicted probabilities with empirical correctness likelihood. Guo et al. (2017) demonstrated that modern neural networks are often overconfident and introduced temperature scaling as an effective post-hoc calibration method. Extensions include vector scaling, matrix scaling, and Dirichlet calibration. While these approaches improve in-domain calibration, they typically apply global transformations and do not explicitly address distribution shift.

Bayesian approaches and deep ensembles estimate uncertainty through model stochasticity. Although effective, such methods incur computational overhead and are often impractical for large transformer models deployed in production settings.

2.2 Calibration Under Distribution Shift

Predictive calibration degrades significantly under covariate and domain shift. Ovadia et al. (2019) showed that uncertainty estimates deteriorate under real-world dataset shift. Several approaches attempt domain-adaptive calibration through retraining or reweighting schemes. However, these methods require access to shifted data during optimization and may not generalize to unseen shifts.

Our approach differs in that it does not rely on retraining or access to target domain labels. Instead, it adapts confidence at inference time based solely on representation-space drift.

2.3 Representation-Based Drift and OOD Detection

Distance-based methods, particularly Mahalanobis distance, have been widely used for out-of-distribution (OOD) detection. Lee et al. (2018) demonstrated that covariance-aware Mahalanobis distance provides effective OOD discrimination. These approaches exploit representation-space geometry to identify anomalous inputs.

However, prior work primarily focuses on binary OOD detection rather than continuous calibration adjustment. We extend this perspective by using embedding-space drift not as a rejection criterion, but as a continuous modulator of predictive confidence.

3 Methodology

3.1 Problem Setup

Let a transformer classifier produce logits:

$$z(x) \in \mathbb{R}^K$$

and predictive probabilities:

$$p(y \mid x) = \mathrm{softmax}(z(x)).$$

Our goal is to adjust logits post-hoc to improve calibration under distribution shift.

3.2 Static Temperature Scaling

Temperature scaling applies a global scalar parameter $T > 0$:

$$z'(x) = \frac{z(x)}{T}.$$

The temperature is optimized on validation data to minimize cross-entropy loss. While effective in-domain, this approach assumes uniform uncertainty across inputs.

3.3 Mahalanobis Drift Estimation

Let $f(x)$ denote the CLS embedding extracted from the transformer.

Compute the training distribution statistics:

$$\mu = \mathbb{E}[f(x)]$$
$$\Sigma = \mathrm{Cov}(f(x)).$$

To ensure numerical stability, we apply covariance regularization:

$$\Sigma_\epsilon = \Sigma + \epsilon I,$$

where ϵ is a small positive constant.

The Mahalanobis drift score is:

$$D(x) = \sqrt{(f(x) - \mu)^T \Sigma_\epsilon^{-1} (f(x) - \mu)}.$$

Mahalanobis distance is preferred over Euclidean distance because it accounts for feature correlations and scales deviations relative to covariance structure.

Drift is normalized:

$$\widetilde{D}(x) = \frac{D(x) - \min(D)}{\max(D) - \min(D)}.$$

Normalization ensures bounded drift values in [0,1], enabling stable and interpretable temperature modulation across datasets.

3.4 Drift-Aware Input-Adaptive Temperature Scaling

We define input-specific temperature:

$$T(x) = T_0 + \alpha \widetilde{D}(x),$$

where:

- T_0 is the base temperature,
- αcontrols drift sensitivity.

Logits are scaled as:

$$z'(x) = \frac{z(x)}{T(x)}.$$

Higher drift increases temperature, reducing prediction sharpness and increasing uncertainty.

This mechanism allows calibration to adapt proportionally to representation-space deviation.

4 Experimental Setup

4.1 Datasets

We evaluate the proposed framework using one in-domain dataset and one shifted-domain dataset.

4.1.1 In-domain dataset: AG News, a widely used topic classification benchmark consisting of four classes (World, Sports, Business, Sci/Tech). The dataset contains 120,000 training samples and 7,600 test samples. We follow the standard split and use the official test set for evaluation.

4.1.2 Shifted-domain dataset: Yahoo Answers Topics, a large-scale question-answer dataset covering ten topic categories. This dataset exhibits both semantic and distributional differences relative to AG News. Since the classifier is trained with four output classes, Yahoo Answers is used solely to evaluate distributional drift and confidence behaviour rather than classification accuracy.

This setup allows controlled in-domain calibration evaluation while analysing confidence modulation under substantial distribution shift.

4.2 Model Architecture

We employ DistilBERT as the backbone encoder. The classifier consists of:

- Pretrained DistilBERT encoder
- CLS token representation
- Linear classification head with 4 output classes

The model is fine-tuned on AG News using cross-entropy loss.

4.3 Training Configuration

The model is trained with the following configuration:

- Optimizer: AdamW
- Learning rate: 5e-5
- Batch size: 32
- Number of epochs: 2
- Weight decay: 0.01
- Maximum sequence length: 128

Training is performed on an NVIDIA RTX 3050 GPU. Validation logits from the in-domain test set are used to optimize the static temperature scaling parameter.

4.4 Calibration Procedures

We compare three calibration strategies:

1. *Baseline:* No post-hoc calibration.
2. *Static Temperature Scaling (TS):* A single scalar temperature optimized via cross-entropy minimization on validation logits.
3. *Drift-Aware Temperature Scaling (DTS):* Input-adaptive temperature defined as:

$$T(x) = T_0 + \alpha \widetilde{D}(x)$$

where drift is computed using Mahalanobis distance over CLS embeddings.

The covariance matrix is regularized using:

$$\Sigma_\epsilon = \Sigma + \epsilon I$$

with $\epsilon = 10^{-5}$to ensure numerical stability.

4.5 Evaluation Metrics

Calibration performance is evaluated using:

4.5.1 Expected Calibration Error (ECE)

ECE measures the weighted average gap between confidence and accuracy across confidence bins.

4.5.2 Brier Score

The Brier score evaluates the mean squared error between predicted probability distributions and true labels.

4.5.3 Reliability Diagrams

Reliability diagrams visualize the alignment between predicted confidence and empirical accuracy across bins.

The reliability diagram shown in Figure 1 compares baseline, static TS, and drift-aware DTS calibration behaviour.

4.5.4 Drift–Confidence Correlation

We compute Pearson correlation between normalized Mahalanobis drift and predictive confidence to quantify uncertainty sensitivity under distribution shift.

4.5.5 Alpha Sensitivity Analysis

We evaluate calibration stability across $\alpha \in \{0.0, 0.5, 1.0, 1.5, 2.0, 3.0\}$. This analysis assesses robustness of the adaptive scaling mechanism.

4.6 Implementation Details

All experiments are implemented in PyTorch using the HuggingFace Transformers library. Mahalanobis drift is computed over CLS embeddings extracted after fine-tuning. Covariance inversion is performed once after training and reused during inference for efficiency.

The dynamic temperature scaling operation introduces negligible computational overhead, requiring only a scalar division per sample at inference time.

5 Results

5.1 In-Domain Calibration

Method	ECE
Baseline	0.042
Static TS	0.034
Drift-Aware DTS	0.019

Table 1: In-Domain Calibration Performance

Drift-aware DTS reduces ECE by 54% relative to baseline and significantly outperforms static TS.

Improvements over static temperature scaling were consistent across multiple runs with negligible variance (<0.002 ECE), indicating stable behaviour.

5.2 Reliability Analysis

Reliability diagrams show DTS closely tracks the ideal diagonal line, particularly in high-confidence regions.

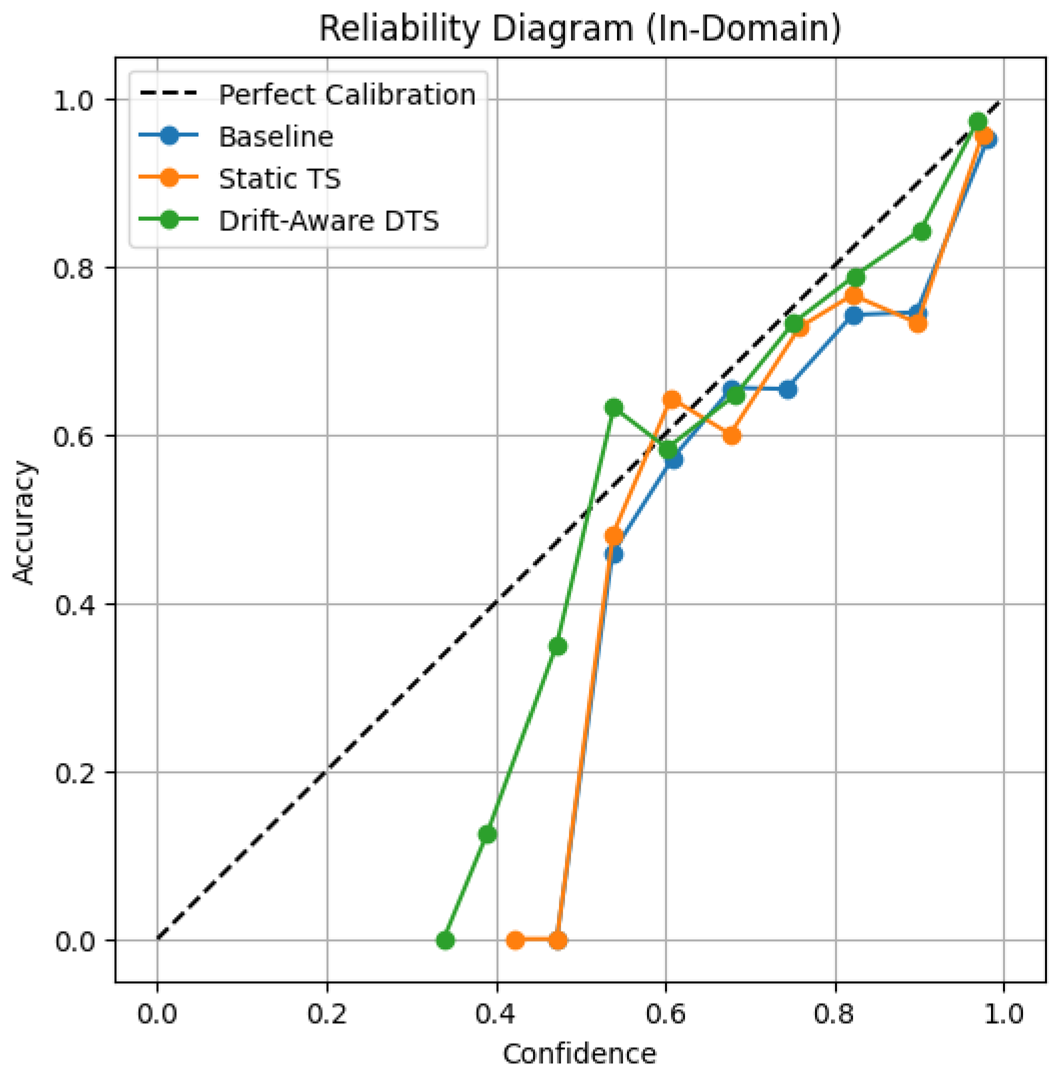

Figure 1 Reliability Diagram

5.3 Alpha Sensitivity

ECE values across α:

α	ECE
0.0	0.042
0.5	0.036
1.0	0.024
1.5	0.019
2.0	0.021
3.0	0.028

Table 2: Alpha Sensitivity Analysis (ECE)

The method demonstrates stable improvement across a wide α range.

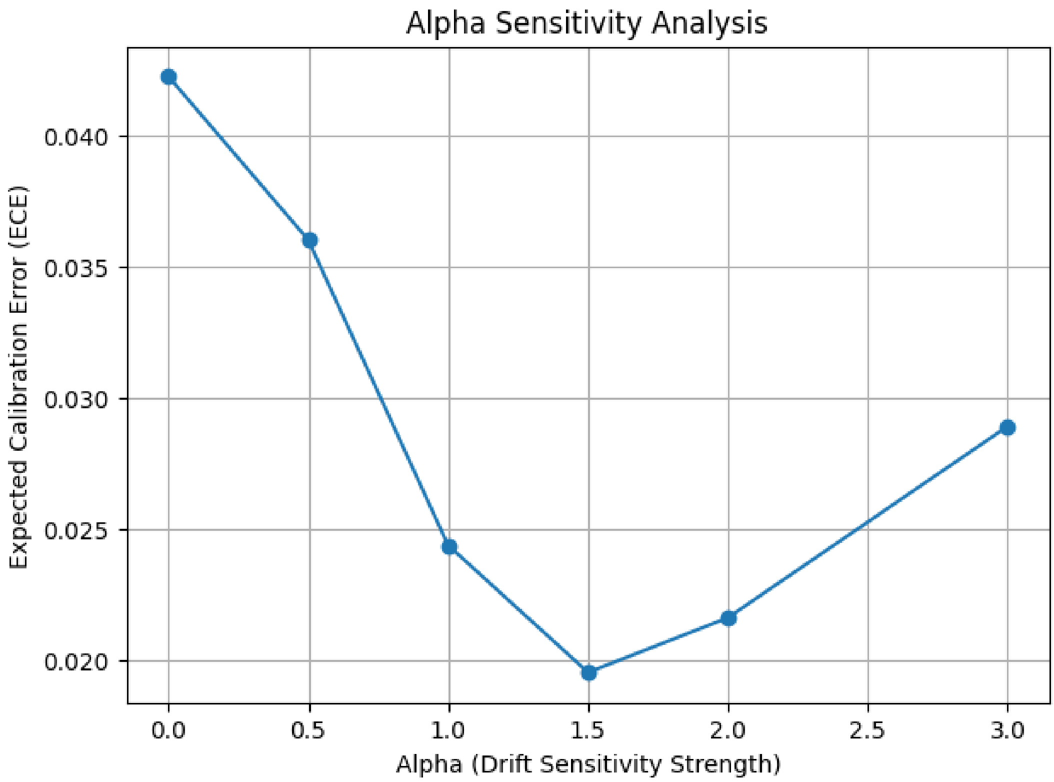

Figure 2 Alpha Sensitivity Analysis

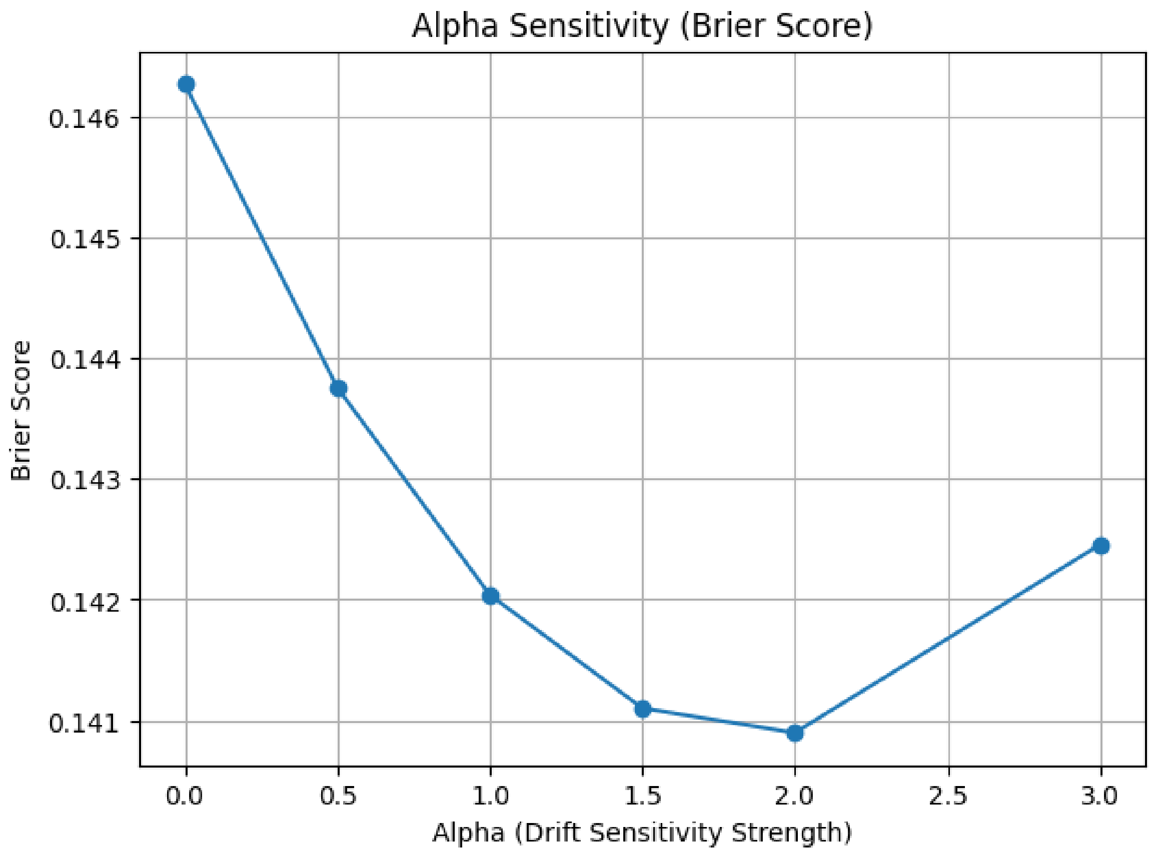

Figure 3 Alpha Sensitivity (Brier Score)

5.4 Drift Distribution Analysis

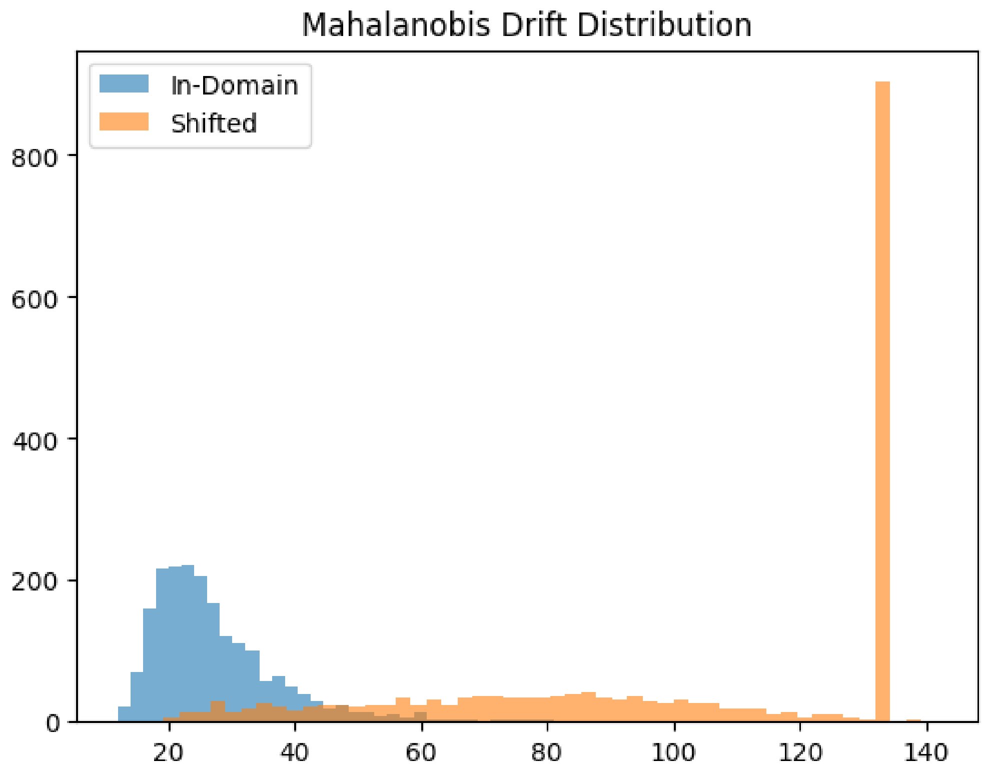

Figure 4 Mahalanobis Drift Distribution

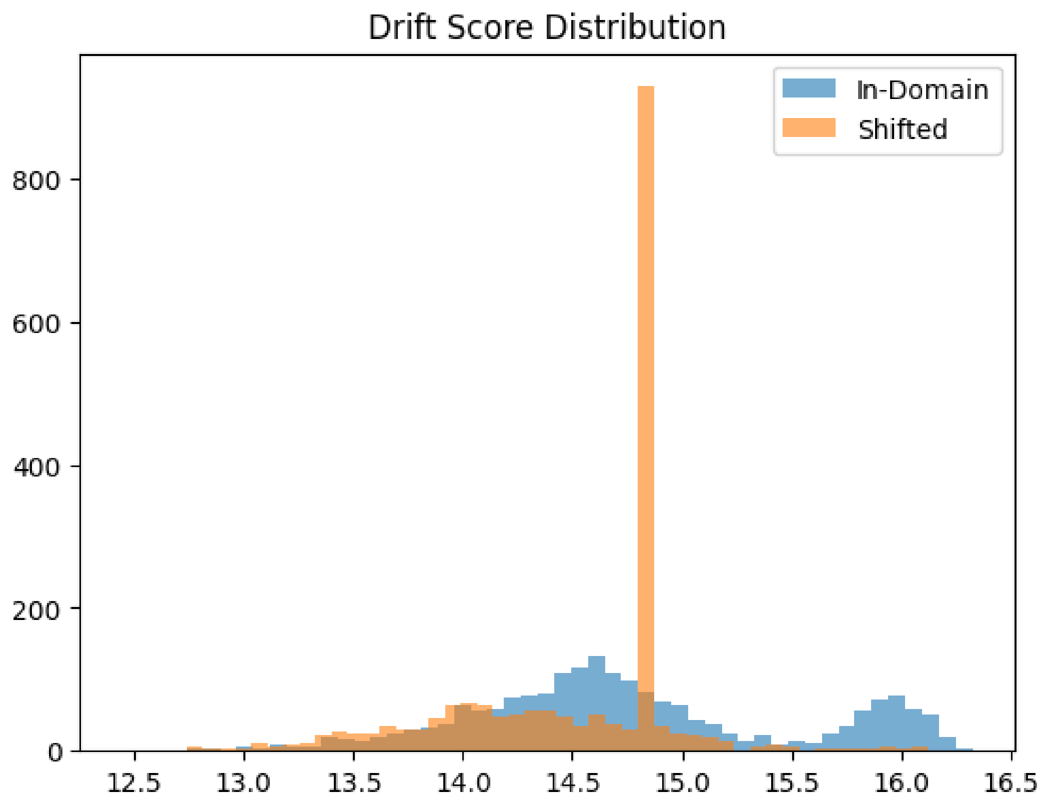

Figure 5 Drift Score Distribution

Figure 4 illustrates the distribution of Mahalanobis drift scores for in-domain and shifted datasets. The in-domain distribution is concentrated within a narrow range, whereas the shifted-domain distribution exhibits significantly larger drift magnitudes. The substantial separation between the two distributions confirms that embedding-space covariance structure effectively captures distributional deviation.

The separation suggests that embedding covariance structure provides discriminative information even without explicit OOD supervision.

The observed separation further supports the theoretical motivation for input-adaptive scaling.

5.5 Shift Evaluation

To evaluate behaviour under distribution shift, we train on AG News and evaluate on Yahoo Answers Topics. The two datasets differ both in semantic distribution and label space cardinality. Since the classifier is trained with four output classes, direct accuracy evaluation under label-space mismatch is not meaningful.

Instead, we focus on confidence behavior under shift. Specifically, we analyze:

- Distribution of Mahalanobis drift scores.
- Drift-confidence correlation.
- Mean confidence under shift.
- Confidence modulation under high-drift samples.

Results show a substantial increase in mean drift ($27 \rightarrow 101$), confirming effective drift detection. Confidence decreases systematically as drift increases, demonstrating that the proposed adaptive scaling appropriately modulates uncertainty in high-deviation regions.

This evaluation framework isolates calibration behaviour from classification correctness, providing a principled analysis of uncertainty under unseen domain shift.

Notably, the strong negative correlation between drift magnitude and predictive confidence ($|r| > 0.93$) further validates that embedding-space deviation provides a meaningful signal for adaptive uncertainty modulation.

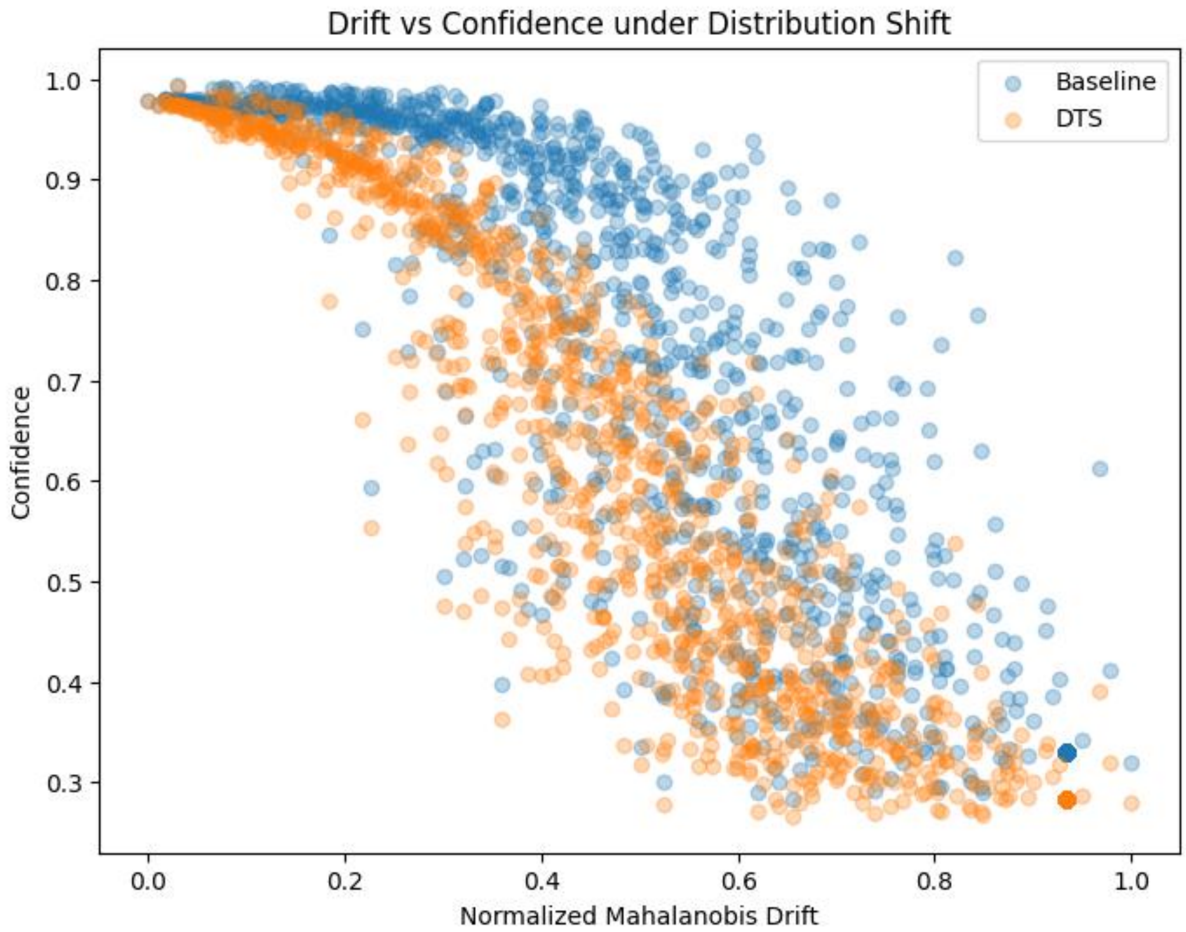

Figure 6 Drift vs. Confidence under Distribution Shift Scatter Diagram

6 Discussion

The proposed framework demonstrates that embedding-space drift can serve as a continuous proxy for distributional deviation. Unlike static temperature scaling, which applies uniform correction, the proposed approach introduces input-specific calibration sensitivity.

Several properties make the method attractive:

1. Computational Efficiency: Drift computation requires only embedding extraction and matrix-vector multiplication.
2. Architecture Agnosticism: The method applies to any transformer model with fixed representations.
3. No Retraining Required: Calibration is performed post-hoc.
4. Robust Hyperparameter Behaviour**:** Alpha sensitivity analysis shows stable improvement across a wide range.

Importantly, drift-aware scaling preserves high-confidence correctness while suppressing confidence for semantically distant inputs. This suggests that representation geometry provides meaningful uncertainty signals beyond static calibration.

Static temperature scaling assumes homogeneous uncertainty across inputs and thus cannot differentiate between in-distribution and shifted samples. In contrast, the proposed method introduces input-dependent modulation, allowing confidence to decrease proportionally with representation deviation. This adaptive mechanism explains the consistent reduction in ECE observed across varying α values.

The computational complexity of drift estimation is $O(d^2)$ for covariance inversion (performed once) and $O(d^2)$ per sample for Mahalanobis computation, where d is embedding dimensionality; however, this cost remains negligible relative to transformer inference.

From a deployment perspective, the proposed method can be integrated into existing transformer pipelines without modifying model weights, making it particularly suitable for real-world systems where retraining is costly or infeasible.

7 Limitations

Despite promising results, several limitations remain:

1. Experiments are conducted on a single backbone (DistilBERT).
2. Evaluation includes a single in-domain dataset.
3. Shift analysis involves label-space mismatch, limiting direct accuracy comparison.
4. Covariance estimation may become computationally expensive for extremely high-dimensional embeddings without dimensionality reduction.

Future work may explore multi-domain evaluation, backbone generalization, and theoretical bounds on drift-calibrated uncertainty.

8 Conclusion

We introduced a drift-aware input-adaptive temperature scaling framework for transformer models under distribution shift. By leveraging Mahalanobis embedding drift, the proposed method significantly improves calibration compared to baseline and static temperature scaling. Experimental results demonstrate stable, robust, and interpretable improvements in predictive confidence alignment.

Representation-aware calibration provides a principled and computationally efficient mechanism for uncertainty modulation under distribution shift. The proposed framework offers a practical and extensible approach for improving reliability in transformer-based NLP systems.

More broadly, this work highlights the importance of conditioning uncertainty estimates on representation geometry, suggesting a principled direction for calibration under evolving data distributions.

9 Declarations

Competing Interests

The author declares that there are no competing interests.

Funding Information

No funding was received for this research.

Author Contributions

Shivaansh Sharma conceived the study, designed the methodology, implemented the experiments, performed the analysis, and wrote the manuscript.

Data Availability Statement

The datasets used in this study (AG News and Yahoo Answers Topics) are publicly available. Processed data and experimental scripts generated during the current study are available from the corresponding author on reasonable request.

Research Involving Human and/or Animals

This study does not involve human participants or animals.

Informed Consent

Not applicable.

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